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A Peridynamic Model for Ductile Fracture of Moderately Thick Plates

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Abstract

This paper presents an effective modeling technique for propagation of mode I cracks in ductile materials using Peridynamics Implemented Finite Element Analysis (PDIFEA) in ABAQUS. Peridynamics is a nonlocal theory which removes the restrictions of fracture modeling in classical continuum mechanics. PDIFEA models of different Compact Tension (CT) specimens are generated using truss elements. The yield stretch of a bond is analytically determined by equating total force across fracture surface in yield state and experimentally measured yield strength of the material. The critical stretch of the bond is related to critical strain energy release rate. A new constitutive model is proposed for the bond failure which depends on yield stretch and critical stretch value. Load versus crack mouth opening displacement results are calculated and validated with the published results in the open literature. Based on the results obtained in this study, it can be indicated that PDIFEA with proposed constitutive model for two-dimensional is capable of capturing ductile damage in CT specimens.

Keywords: peridynamics; fracture mechanics; finite element analysis; nonlocal; failure

Nomenclature

PD	Peridynamics
FEA	Finite Element Analysis
PDIFEA	Peridynamic Theory Implemented in Finite Element Analysis
CCM	Classical Continuum Mechanics

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1. Introduction

High strength steels and aluminum alloys are strong candidates for lightweight structures in automotive and aerospace industries because of their high ductility and fracture behavior. These materials have stable crack growth phase unlike composite materials. Crack propagation is resisted by the plastic region ahead of the crack tip in ductile materials. Thus there is a stable crack growth phase before ultimate failure (Kumar et al., 2014). In this phase, mechanical properties of the structure may degrade substantially. Thus numerical estimation of crack growth is critical in the design phase of the components from the aspects of fracture mechanics.

Fracture mechanics concepts are developed in order to predict propagation of cracks in brittle materials. Griffith (1921) solved singular stress fields at the crack tip for glass which lead to Linear Elastic Fracture Mechanics (LEFM) concept. LEFM assumes small scale yielding zone at the crack tip where fracture process zone is inside singularity dominated zone. LEFM is not applicable when excessive plastic deformation is present at the crack tip. For such cases Elastic Plastic Fracture Mechanics (EPFM) was developed (Rice, 1968). LEFM and EPFM concepts focus on propagation of pre-existing cracks rather than the nucleation of new cracks (Madenci and Oterkus, 2014). Note that, an external criterion such as fracture toughness or critical energy release rate is required for crack growth analyses in fracture mechanics concepts.

Over the years, fracture mechanics concepts are integrated in robust Finite Element Tools. Finite Element Analyses (FEA) of crack propagation requires special elements at the crack tips to correctly model the singular behavior. Extended Finite Element Method (XFEM) is one of those methods that can correctly model cracks using enrichment functions (Belytschko and Black, 1999; Melenk and Babuška, 1996). However, XFEM also requires an external failure criteria and incorporation of discontinuity as additional terms in the displacement formulation (Song et al., 2007).

The difficulty of crack modeling in classical FEA comes from its governing equations which require spatial derivatives. Spatial derivatives are not defined at crack tips by definition. As a remedy, Silling proposed a new nonlocal theory of continuum mechanics called Peridynamic (PD) theory (Silling and Askari, 2005; Silling, 2000). PD theory employs integro-differential equations in its governing equations which remains valid at discontinuities. Thus, a special treatment at the crack tip is not necessary and crack initiation and crack growth can be analyzed without requiring an external criteria (Oterkus and Madenci, 2012a). In PD theory, a material point interacts with its neighbors within a finite radius. By this aspect, PD theory can be treated as an upscaled version of molecular dynamics (Seleson et al., 2009).

PD theory can be divided into two subcategories: bond based formulation and state based formulation (Madenci and Oterkus, 2014). Bond based formulation assumes pairwise interaction between material points whereas state based formulation also accounts for indirect interactions between material points (Silling et al., 2007). In bond based formulation, Poisson ratio of the material is limited to 1/3 in plane stress problems and 1/4 in plane strain problems due to pairwise interactions (Madenci and Oterkus, 2014).

There have been various studies on the application of peridynamics to mechanics problems in the literature. Oterkus et al. (2012b) investigated damage in a stiffened curved composite panel by coupling PD theory with FEA. Oterkus and Madenci (2012b) derived PD bond constants for a fiber reinforced lamina in terms of engineering constants E_{11} , E_{22} , G_{12} and ν_{12} . Oterkus et al. (2012a) studied several problems including longitudinal vibration of a bar, propagation of pre-existing crack in a plate, and crack initiation and growth in a plate by driving governing equations of Peridynamics based on principle of virtual work. Taylor and Steigmann (2013) derived a two dimensional PD plate theory which is able to model fractures due to bending. Dipasquale et al. (2014) proposed an adaptive grid refinement method in two dimensional PD problems based on damage and energy concentration which increases the computational efficiency. Ha and Bobaru (2010) captured dynamic fracture and crack branching events in brittle materials using PD theory by validating their results with experiments from the literature. Silling and Bobaru (2005) studied dynamic fracture of nanofiber networks and membranes and captured transition from mode III fracture to mode I fracture in membranes. Hu et al. (2013) worked on numerical simulation of impact on two layer glass laminate with polycarbonate back plate and they validated their numerical results with experiments. Hu et al. (2012) proposed a homogenized PD model for brittle fracture in fiber reinforced composites. Wu and Ren (2015) proposed a new formulation to model ductile material failures during machining process by combining local/nonlocal gradient

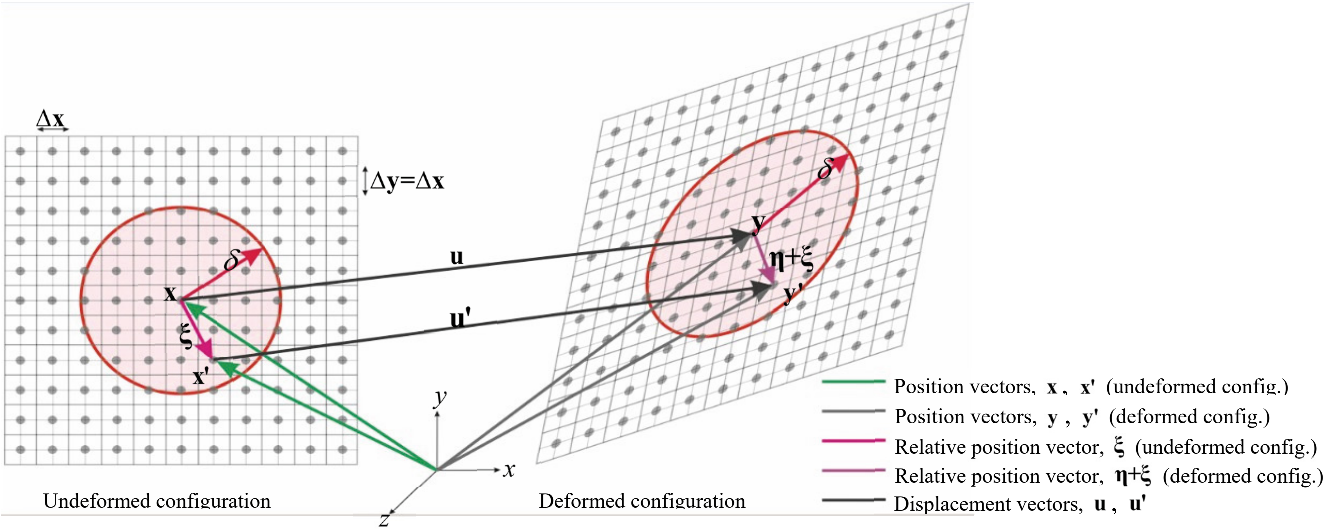


Fig. 1. Kinematics of material points from undeformed configuration to deformed configuration

approximations with PD force vector state. Liu and Hong (2012) implemented a three-dimensional PD code in GPU to analyze brittle and ductile solids and they proposed a constitutive relation to represent ductile behavior of solids. Lee et al. (2015) developed a numerical scheme to model contact between non PD domain and PD domain for impact simulations. They also employed a constitutive relation at bond level to approximate macroscopic ductile behavior. Madenci and Oterkus (2016) developed an ordinary state based PD plasticity model with Von-Mises yield criteria and isotropic hardening.

In the literature, peridynamics is mostly applied to materials exhibiting brittle fracture behavior. In this study, fracture mechanics problems with excessive plastic deformations are investigated using bond based Peridynamics Implemented Finite Element Analysis (PDIFEA) method. To approximate ductile behavior of the material, a new constitutive relation for bonds is proposed. Mathematical background of this study is explained in Section 2. Constitutive model and the material constants are derived in Section 3. Section 4 explains the implementation of PD theory in FEA. The solutions of stable crack growth in aluminum CT specimens and steel triple bend specimen are detailed in Section 5. Finally, the conclusions drawn from this study are discussed in Sections 6.

2. Peridynamic theory

In Classical Continuum Mechanics (CCM), deformation of a body subjected to external loads can be calculated by treating the body as continua. In CCM, it is assumed that a continuous medium has infinite number of infinitesimal volumes which interact only with their immediate neighbors. Equation of motion in CCM can be written as:

$$-\rho(\mathbf{x})\ddot{\mathbf{u}} + \nabla.\boldsymbol{\sigma} + \mathbf{b}(\mathbf{x},t) = 0, \tag{1}$$

where $\rho(\mathbf{x})$, $\ddot{\mathbf{u}}$ and $\mathbf{b}(\mathbf{x},t)$ denote mass density, spatial acceleration and body force density vector of the material point $\mathbf{x}$, respectively. The spatial derivatives of stress tensor $\boldsymbol{\sigma}$ required in Eq. (1) are not valid at discontinuities. In PD theory, equation of motion does not require spatial derivatives and can be written as:

$$\rho(\mathbf{x})\ddot{\mathbf{u}}(\mathbf{x},t) = \int_H \mathbf{f}(\mathbf{u}' - \mathbf{u}, \mathbf{x}' - \mathbf{x})dH + \mathbf{b}(\mathbf{x},t)\,. \tag{2}$$

In Eq. (2), H is the horizon of material point $\mathbf{x}$ with a radius of δ as illustrated in Fig. 1, ρ is the mass density, $\mathbf{u}$ and $\mathbf{b}$ denotes the displacement and body force vector fields, respectively. Note that $\mathbf{f}(\mathbf{u}(\mathbf{x}')-\mathbf{u}(\mathbf{x}),\mathbf{x}'-\mathbf{x})$ is the pairwise force vector that the material point $\mathbf{x}$ exerts on the material point $\mathbf{x}'$. It has been shown that, PD theory converges to classical theory of elasticity, when the length of the horizon goes to zero (Silling and Lehoucq, 2008).

In PD theory, force interactions between material points can be modelled using PD bonds. As represented in Fig. 1, a material point has force interactions with all its neighbors in its horizon. Note that δ is taken to be $3\Delta x$ (see Section 4 for the details) in Fig. 1 which corresponds to 28 material points within the horizon.

Magnitude of force between two material points $\mathbf{x}$ and $\mathbf{x}'$ is a function of initial distance ($\boldsymbol{\xi}$) and relative displacement $(\boldsymbol{\eta})$ (see Fig. 1) which can be defined as:

$$\boldsymbol{\xi} = \mathbf{x}' - \mathbf{x}\,,\ \ \boldsymbol{\eta} = \mathbf{u}(\mathbf{x}',t) - \mathbf{u}(\mathbf{x},t)\,,\ \ \boldsymbol{\eta} + \boldsymbol{\xi} = \mathbf{y}' - \mathbf{y}\,, \tag{3a,b,c}$$

where $\mathbf{y}' - \mathbf{y}$ is the deformed form of initial bond vector, $\boldsymbol{\xi}$. In bond based PD formulation, force density vector between material points is given by (Madenci and Oterkus, 2014)

$$\mathbf{f}(\boldsymbol{\eta},\boldsymbol{\xi}) = \frac{\boldsymbol{\xi}+\boldsymbol{\eta}}{|\boldsymbol{\xi}+\boldsymbol{\eta}|}cs\,,\ \ s = \frac{|\boldsymbol{\xi}+\boldsymbol{\eta}| - |\boldsymbol{\xi}|}{|\boldsymbol{\xi}|}\,, \tag{4a,b}$$

where c is material constant and s is the magnitude of stretch.

For micro-elastic materials, the relation between the force density vector and the micropotential, $w(\boldsymbol{\eta},\boldsymbol{\xi})$, is defined by Silling (2000) as

$$\mathbf{f}(\boldsymbol{\eta},\boldsymbol{\xi}) = \frac{\partial}{\partial \boldsymbol{\eta}} w(\boldsymbol{\eta},\boldsymbol{\xi})\,. \tag{5}$$

PD strain energy density can be found by integrating the micropotential with respect to horizon:

$$W^{PD} = \frac{1}{2}\int_H w(\boldsymbol{\eta},\boldsymbol{\xi})\,dH\,. \tag{6}$$

3. Constitutive Model and Material Constants

In real engineering problems, the mechanical behavior of materials is approximated by using constitutive models which are also known as material models. In general, those models have several constants such as elastic modulus and Poisson's ratio which characterize the type of material. PD material constants should be defined in terms of experimentally measureable engineering constants for modeling real materials using PD theory. At bond level, elastic properties can be represented with micromodulus, c (see Eq. (4)), which defines the modulus of a bond. The micromodulus for a two dimensional plate can be determined by comparing the strain energy densities in PD theory and CCM under isotropic expansion loading $(s = \zeta)$ as shown in Fig. 2.

Micropotential of a PD bond for the plate under isotropic expansion loading can be expressed as:

$$w = \frac{1}{2}c\zeta^2\xi\,, \tag{7}$$

where ζ is applied stretch for a PD bond.

Using Eq. (6) and Eq. (7), PD strain energy density can be written as:

$$W^{PD}_{is.\exp} = \frac{1}{2}t\int_0^{2\pi}\int_0^{\delta}\left(\frac{c\zeta^2\xi}{2}\right)\xi\,d\xi\,d\theta = \frac{\pi t\delta^3 c\zeta^2}{6}\,. \tag{8}$$

In CCM, strain energy density of a plate under isotropic expansion is given as:

$$W^{CCM}_{is.\exp} = \frac{E}{(1+\nu)(1-2\nu)}\zeta^2\,. \tag{9}$$

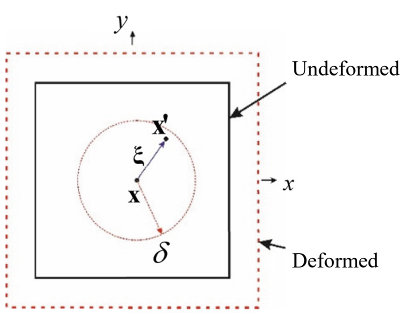


Fig. 2. Two-dimensional plate under isotropic expansion loading

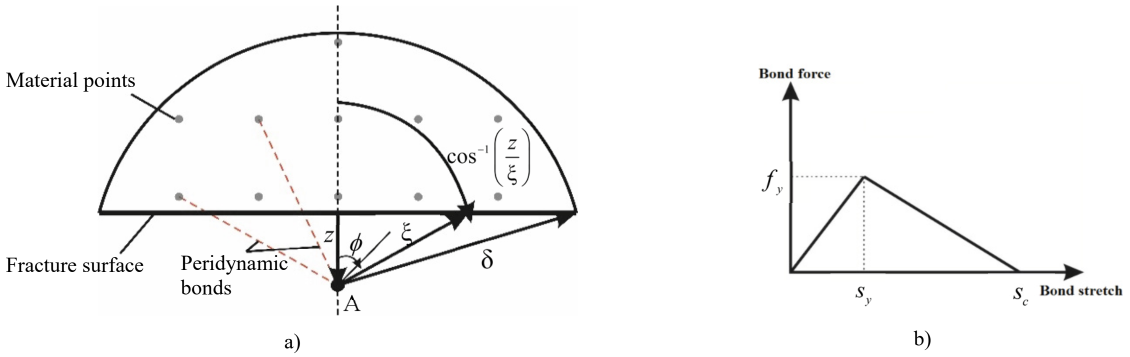


Fig. 3. a) Two-dimensional integration domain across fracture surface (only some of peridynamic bonds are indicated explicitly for clarity) b) Constitutive model for a peridynamic bond

Eq. (9) is written under plane strain condition since compact tension specimens in this study are moderately thick plates. Micromodulus for two-dimensional isotropic plates can be obtained by equating strain energy densities (Eq. (8) to Eq. (9)) in CCM and PD theory. The solution leads to the following relations

$$c_{2D} = \frac{48E}{5\pi t\delta^3}, \quad \nu_{2D} = \frac{1}{4}, \tag{10a,b}$$

where t is plate thickness. It should be noted that Poisson's ratio, ν, is taken as 1/4 due to constraints of bond based PD formulation on Poisson's Ratio (Gerstle et al., 2005).

3.1. Failure model

In this study, a micro-plastic PD constitutive model is employed to account for the yielding of the material. A bilinear constitutive law for PD bonds is proposed as illustrated in Fig. 3b. At bond level, this relation is linear elastic until the yield point and then softens linearly. However, this relation can capture nonlinear macroscopic behavior since all of the bonds do not yield at the same time (Macek and Silling, 2007).

Yield stretch of bonds can be related to yield strength, σ_y, of the material with the assumption that all bonds yields when the material reaches its yield strength. For three-dimensional structures, definition of yield stretch which is related to ultimate tensile strength, σ_{ult}, is given by Macek and Silling (2007). In this paper, we propose a two-dimensional formulation for the yield stretch, s_y. A relation between yield stress and force density at yield, f_y, can be obtained by integrating the PD force density within the horizon (see Fig. 3a)

$$\sigma_y = 2t\int_0^{\delta}\left\{\int_z^{\delta}\int_0^{\cos^{-1}(z/\xi)} f_y \xi\cos(\phi)\,d\phi\, d\xi\right\}dz\,, \quad f_y = cs_y\,. \tag{11a,b}$$

Definition of yield stretch in two-dimensions, s_y^{2D}, can be obtained as

$$s_y^{2D} = \frac{5\sigma_y}{8E}\,. \tag{12}$$

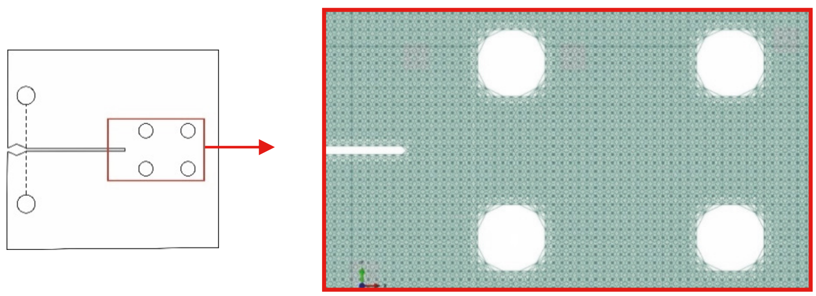


Fig. 4. PDIFEA mesh

The material failure is considered by bond breakage when the stretch in the bond exceeds a critical stretch value, s_c . Critical stretch value can be determined by calculating the area under stretch vs. force density curve (see Fig. 3b). This can be related to critical strain energy release rate G_{IC} as:

$$G_{\mathrm{IC}} = \frac{1}{2}\left(f_y s_c \left|\mathbf{x'}-\mathbf{x}\right|\right) V^{4/3}, \tag{13}$$

where V is volume of a material point. A critical stretch value for two dimensional structures, s_c, can be written by substitution of Eq. (11b) into Eq. (13) as

$$s_c = \frac{2G_{\mathrm{IC}}}{c s_y V^{4/3}\left|\mathbf{x'}-\mathbf{x}\right|}. \tag{14}$$

4. Implementation of Peridynamic Theory in FEA

Peridynamics is based on modeling the interactions of material points within a finite horizon. Even though it is known as a “mesh-free method”, mesh structures made of truss elements or linear springs can be used in a finite element code to represent PD bonds. Macek and Silling (2007) used commercial finite element code ABAQUS to simulate brittle central cracked plate and they validated their results with PD code EMU.

In this study, a similar procedure to Macek and Silling (2007) was followed utilizing the commercial FE code ABAQUS/Explicit. Analysis inputs were prepared by using MATLAB scripts using the geometry of the specimens given by Kumar et al. (2014). PD bonds are modeled using truss elements. Three dimensional mechanical truss elements (T3D2) in the ABAQUS element library were generated by linking each node to its neighbor nodes within the horizon, δ . Nodes and elements in PDIFEA model represent material points and pairwise interactions between material points, respectively, in a bond based PD model. An example of the PDIFEA mesh is illustrated in Fig. 4.

In the literature, there are convergence studies on the radius of the horizon (Gerstle et al., 2007; Ha and Bobaru, 2010; Silling and Askari, 2005). It is commonly suggested that for macroscopic simulations, a horizon length of $\delta = 3\Delta x$ gives accurate results. Larger horizon radius increases computational cost without improving the results. Thus, in this study PDIFEA models are generated using $\delta = 3\Delta x$.

PD force formulation can be adapted to FEA by relating the force in the truss element to PD force density as follows (Macek and Silling, 2007)

$$A_t = V^{(2/3)},\ E_t = cV^{(4/3)}, \tag{15a,b}$$

where c is the bond constant.

For the truss elements, bond stretch s is identical to strain ε (Macek and Silling, 2007). To model degradation and failure of the bond in the constitutive model (see Fig. 3b), yield stress, σ_y^{truss} , and fracture strain, ε_f^{truss} , of each truss can be determined as:

$$\sigma_y^{truss} = s_y E_t,\ \varepsilon_f^{truss} = s_c, \tag{16a,b}$$

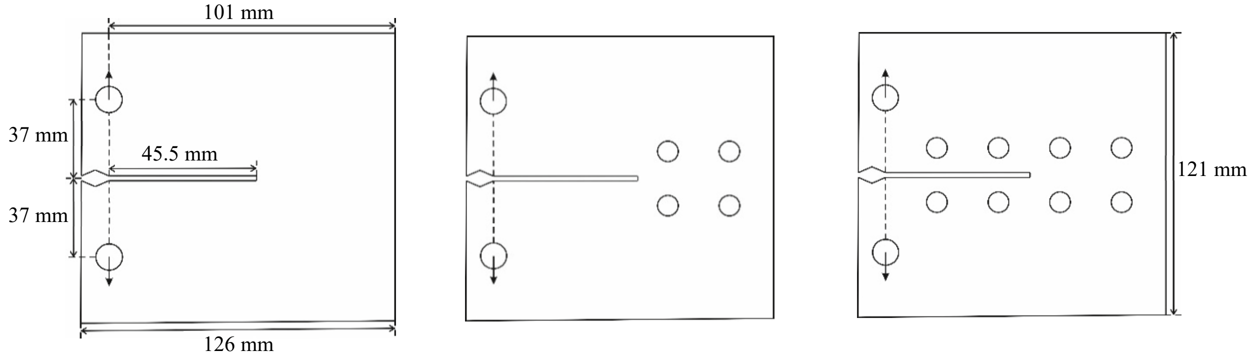
where s_c depends on the truss length.

The stability for explicit procedure is related to time required for a stress wave to cross the smallest element. To provide numerical stability in ABAQUS/Explicit, time step should be less than the ratio of smallest characteristic length of truss element to the dilatational wave speed of the material as follows (DS Simulia, 2012)

$$\Delta t \le \frac{L^{truss}}{c_{dilatational}},\ c_{dilatational} = \sqrt{E_t/\rho}. \tag{17a,b}$$

Note that, wave speed magnitude for PDIFEA models is very high since truss weights are assumed to be nearly zero. When compared to very slow loading rate of PDIFEA models, there may be undesired results such as propagation of stress wave through the model in case of an instantaneous loading. Thus quasi-static procedure was applied to PDIFEA models to avoid instable solutions. When applying the boundary conditions, smooth step amplitude curve is defined.

5. Results



a) b) c)

Fig. 5. Compact tension specimens a) with no circular cutouts b) 4 circular cutouts c) 8 circular cutouts

PDIFEA models of Compact Tension (CT) specimens made of aluminum with three different geometries (see Fig. 5) are generated using the method developed in this study. In order to compare the results obtained from this study to the experimental results of Kumar et al. (2014), the problem is solved with the same boundary conditions and the dimensions as shown in Fig. 5. For these specimens, ductile crack growth in plane strain condition is used and the thickness of the CT specimens is taken to be 10 mm.

PD constants of D16AT aluminum alloy is derived from the macroscopic material parameters and are given in Table 1.

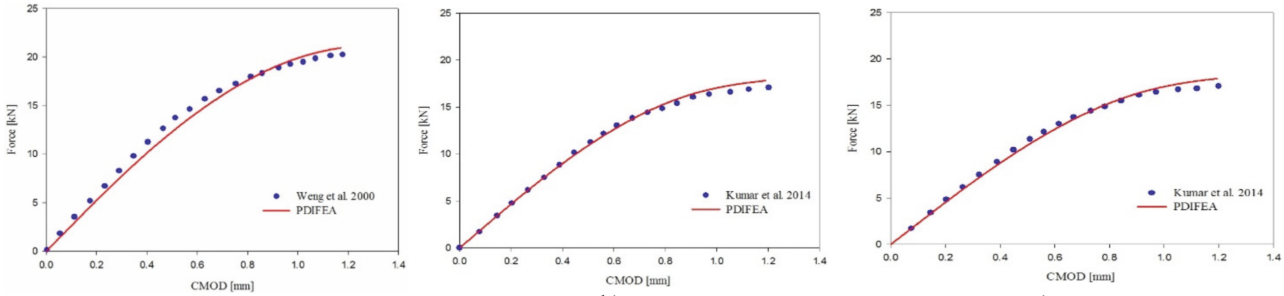
In CT specimens, displacement boundary conditions are applied through the holes, 37 mm away from the pre-existing crack as shown in Fig. 5. Crack mouth opening displacement (CMOD) is measured just above and below the crack. First, load vs. CMOD of CT specimen without holes is calculated with PDIFEA approach. In Fig.6a, the comparison between the PDIFEA result and experimental result of Weng and Sun (2000) for CT specimen without holes is compared. There is a good agreement with the experimental results given by Weng and Sun (2000). Next, load vs. CMOD graphs of CT specimens with 4 holes and 8 holes are calculated (see Fig. 6b and 6c). Obtained results for CT specimens are in good agreement with numerical study of Kumar et al. (2014).

Table 1. Material Properties of specimens (Kumar et al., 2014)

Material	E (GPa)	ν	σ_y (MPa)	G_{IC} (MPa.mm)
D16AT Aluminum	72.6	0.30	304	16.16

6. Conclusion

PDIFEA analyses of CT specimens and three point bending specimen in plane strain condition are performed in this study by using commercial FEA code ABAQUS. A new constitutive model is derived for PD bonds. Two-dimensional formulation of this model is used in CT specimens.



a) b) c)

Fig. 6. Comparison of force-CMOD curves for CT specimens a) with no circular cutouts b) 4 circular cutouts c) 8 circular cutouts

CT specimens with and without holes are analyzed using PDIFEA. The solutions of CT specimens are validated by comparing load vs CMOD curves with the ones from literature. Based on CT results, it can be concluded that holes behind the crack tip reduces the load carrying capacity of the specimens. It can also be noted that, holes in front of the crack tip has no influence on the load carrying capacity by comparing CT specimens with 4 and 8 holes.

Based on those results, it can be indicated that PDIFEA with proposed constitutive two-dimensional is capable of capturing ductile damage in CT. PDIFEA approach is also validated with the results in the open literature.

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